

CHAPTER 6

(6-1) DIAMETER OF SHAFT = 50 mm ; LENGTH OF SHAFT = 1.5 m
TWISTING MOMENT = 1100 N·m

$$\begin{aligned}\text{SHEAR STRESS } \tau_{\theta z} &= \frac{M_t r}{I_z} ; \quad I_z = \frac{\pi d^4}{32} = \frac{\pi r^4}{2} \\ &= \frac{2 M_t}{\pi r^3} = \frac{2 \times 1.1 \times 10^3}{\pi (25)^3 \times 10^{-9}} = 44.8 \text{ MN/m}^2\end{aligned}$$

FROM TABLE 5.1 $G = 80 \text{ GN/m}^2$

$$\text{ANGLE OF TWIST } \phi = \frac{M_t L}{G I_z} = \frac{1100 \times 1.5 \times 2}{80 \times 10^9 \times \pi (25)^4 \times 10^{-12}} = 3.36 \times 10^{-2} \text{ rad}$$

(6-2) LET INSIDE DIAMETER OF HOLLOW STEEL SHAFT = r
OUTSIDE " " " " " = R
LENGTH OF " " " " " = 2.5 m
TWISTING MOMENT $M_t = 25 \text{ kN}\cdot\text{m}$

MAXIMUM ALLOWABLE SHEARING STRESS $\tau_{\theta z \max} = 82 \text{ MN/m}^2$

" " ANGLE OF TWIST $\phi_{\max} = 2.0^\circ = 0.035 \text{ rad}$

$$I_z = \frac{\pi}{2} (R^4 - r^4) = \frac{M_t L}{G \phi} = \frac{25 \times 2.5 \times 10^3}{80 \times 10^9 \times 3.5 \times 10^{-2}} = 2.232 \times 10^{-5}$$

$$\text{FROM SHEARING STRESS EQ } R = \frac{\tau_{\theta z \max} I_z}{M_t} = \frac{82 \times 10^6 \times 2.232 \times 10^{-5}}{25 \times 10^3}$$

$$= 73.2 \text{ mm}$$

$$\frac{\pi}{2} [(73.2)^4 - r^4] = 2.232 \times 10^7$$

$$r = 61.7 \text{ mm}$$

$$R = 73.2 \text{ mm}$$

$$r = 61.7 \text{ mm}$$

6.3

Symmetry argument of § 6.2 stands: radii remain radii
As in § 6.3, $\sigma_r = \sigma_\theta = \sigma_z = \tau_{r\theta} = \tau_{rz} = 0$.

$$\tau_{\theta z} = G_1 r \frac{d\phi}{dz} \quad 0 < r < r_i$$

$$\tau_{\theta z} = G_2 r \frac{d\phi}{dz} \quad r_i < r < r_o$$

$$\begin{aligned} \text{As in § 6.4, } M_t &= \int_A r \tau_{\theta z} dA \\ &= \frac{d\phi}{dz} \left\{ 2\pi G_1 \int_0^{r_i} r^3 dr + 2\pi G_2 \int_{r_i}^{r_o} r^3 dr \right\} \\ &= \frac{\pi}{2} \frac{d\phi}{dz} \left\{ (G_1 - G_2) r_i^4 + G_1 r_o^4 \right\} \end{aligned}$$

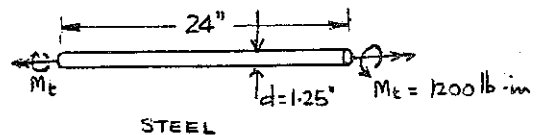
This may be written $M_t = \frac{d\phi}{dz} \{ G_1 I_1 + G_2 I_2 \}$ where $\begin{cases} I_1 = \frac{\pi}{2} r_i^4 \\ I_2 = \frac{\pi}{2} (r_o^4 - r_i^4) \end{cases}$

Total angle of twist $\phi = \frac{M_t L}{G_1 I_1 + G_2 I_2}$

$$\tau_{\theta z} = G_1 r \frac{d\phi}{dz} = \frac{G_1 M_t r}{G_1 I_1 + G_2 I_2}, \quad 0 < r < r_i$$

$$\tau_{\theta z} = G_2 r \frac{d\phi}{dz} = \frac{G_2 M_t r}{G_1 I_1 + G_2 I_2}, \quad r_i < r < r_o.$$

6.4 $(\tau_{\theta z})_{\max} = \frac{M_t r_o}{I_2} = \frac{2 M_t}{\pi r_o^3}$
 $= \frac{2 \times 1200}{\pi \times \left(\frac{1.25}{2}\right)^3} = \underline{\underline{3140 \text{ psi}}}$



$$\begin{aligned} \phi &= \frac{M_t L}{G I_2} = \frac{2 M_t L}{\pi G r_o^4} = \frac{2 \times 1200 \times 24}{\pi \times 11.5 \times 10^6 \times \left(\frac{1.25}{2}\right)^4} = 0.0105 \text{ radians} \\ &= \underline{\underline{0.60 \text{ degrees}}} \end{aligned}$$

6.5 For 1020 CR, $Y \approx 86,000 \text{ psi}$
 $G \approx 11.5 \times 10^6 \text{ psi}$

Let the inner dia. be r "

$$I_z = \frac{\pi}{2} r_o^4 \left[1 - \left(\frac{r}{r_o} \right)^4 \right] = \frac{\pi}{2} (1 - r^4) \text{ in}^4$$

where $r_o = 1$ "

$(\tau_{\theta z})_{\max} = G r \frac{d\phi}{dz}$ should be less than $\frac{Y}{2}$, using Tresca criterion
 (or less than $\frac{Y}{\sqrt{3}}$ using von Mises criterion)

$$\text{or } G r \frac{d\phi}{dz} = \frac{M_t r_o}{I_z} \leq \frac{86,000}{2}$$

The equality sign gives the min. possible I_z , i.e., the max. possible r .

(a) $M_t = 3 \times 10^4 \text{ lb-in.}$

$$\therefore \frac{3 \times 10^4 \times 1}{\frac{\pi}{2} (1 - r^4)} = 43,000 \quad \text{giving } \underline{r_{\max} = 0.86"}$$

(b) $M_t = 6 \times 10^4 \text{ lb-in.}$

$$\therefore \frac{6 \times 10^4 \times 1}{\frac{\pi}{2} (1 - r^4)} = 43,000, \quad \text{giving } \underline{r_{\max} = 0.575"}$$

6.6 Draw a free-body of the upper shaft:

Equilibrium:

$$\sum M_{\text{axis}} = 0 : M_{t2} = F_{DT} \times 75 \quad \text{--- (1)}$$

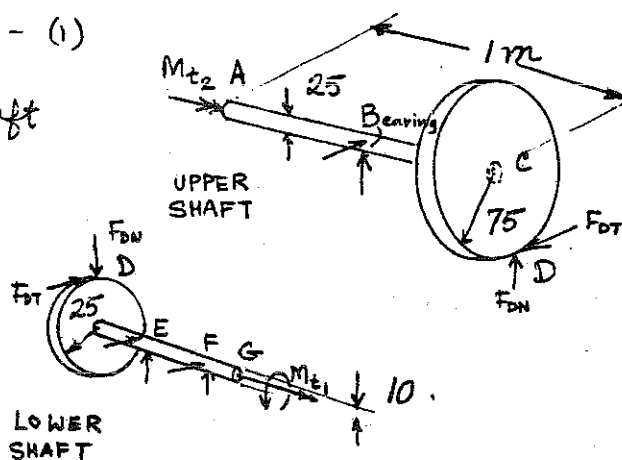
Draw a free-body of the lower shaft

Equilibrium:

$$\sum M_{\text{axis}} = 0 : M_{t1} = F_{DT} \times 25 \quad \text{--- (2)}$$

$$M_{t1} = 10 \text{ N}\cdot\text{m}$$

$$\text{From (1) and (2), } M_{t2} = 30 \text{ N}\cdot\text{m}$$



6.6 CONT

GEOMETRIC COMPATIBILITY: $\phi_G = 3\phi_{AD} + \phi_{DG}$

$$\phi_{AD} = \left(\frac{2M_{t2}L_2}{\pi G r_2^4} \right) = \frac{2 \times 30 \times 1}{\pi \times 80 \times 10^9 \times (12.5)^4 \times 10^{-12}} \text{ (Rad)} = 0.560^\circ$$

$$\phi_{DG} = \frac{2 \times 10 \times 0.67}{\pi \times 80 \times 10^9 \times 5^4 \times 10^{-12}} \text{ Rad} = 4.88^\circ$$

$$\phi_C = 6.56^\circ \text{ COUNTERCLOCKWISE}$$

6.7 Refer to solution of prob. 6.6.

Shaft AD: $(\tau_{\theta z})_{\max} = \frac{2M_{t2}}{\pi r_{o2}^3}$

Shaft DG: $(\tau_{\theta z})_{\max} = \frac{2M_{t1}}{\pi r_{o1}^3}$

$$\frac{(\tau_{\theta z})_{\max, DG}}{(\tau_{\theta z})_{\max, AD}} = \frac{M_{t1}}{M_{t2}} \cdot \frac{r_{o2}^3}{r_{o1}^3} = \frac{1}{3} \left(\frac{12.5}{5} \right)^3 = 5.21$$

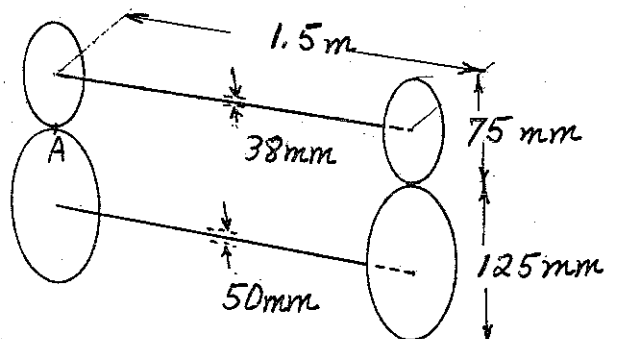
$$(\tau_{\theta z})_{\max, DG} \text{ IS THE LARGEST STRESS} = \frac{2M_{t1}}{\pi (5)^3 \times 10^{-9}} = 275 \times 10^6 \text{ N/m}^2$$

$$M_{t1} = 54 \text{ N}\cdot\text{m}$$

6.8 After assembly, let the lower gear rotate at one end through θ .

Then for geometric compatibility, the upper gear must rotate through $\frac{5}{3}\theta$

Before assembly, the lower gear



6-8 CONT

HAD BEEN ROTATED THROUGH $3^\circ = \frac{3\pi}{180}$ Radians

Thus, THE FINAL ANGLE OF TWIST IN LOWER SHAFT $= \left(\frac{3\pi}{180} - \theta\right)$ Rad

Let $\bar{F}_A$ = TANGENTIAL FORCE INTERACTION AT A

THEN, FOR LOWER SHAFT,

$$\frac{\bar{F}_A 62.5}{L_L} = \frac{\left(\frac{3\pi}{180} - \theta\right) G_L I_L}{L_L} = \left(\frac{3\pi}{180} - \theta\right) G \frac{\pi}{2} \frac{(r_L)^4}{L_L}$$

$$\bar{F}_A 37.5 = \theta G \frac{\pi}{2} \frac{r_u^4}{L_u} \quad G = 80 \text{ GN/m}^2$$

SOLVE ABOVE $\theta = 1.92^\circ \quad \bar{F}_A = 9.756 \text{ kN}$

$$(\tau_{\theta z})_{\text{max, upper}} = 33.8 \text{ MN/m}^2$$

$$(\tau_{\theta z})_{\text{max, lower}} = 24.8 \text{ MN/m}^2$$

6.9 Let F_{AT} , F_{BT} be the tangential force interactions at A & B.

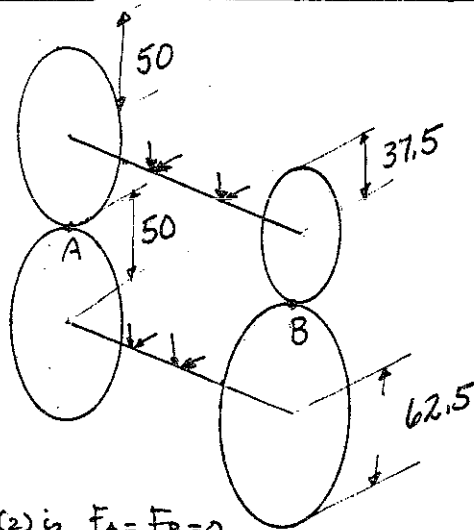
Then, for equilibrium, we must have, taking moments about shaft axes:

Upper Shaft:

$$F_A 50 + F_B 37.5 = 0 \quad (1)$$

Lower Shaft

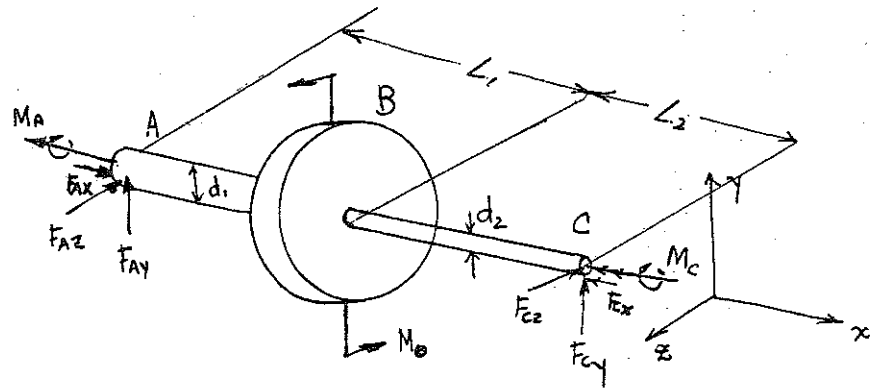
$$F_A 50 + F_B 62.5 = 0 \quad (2)$$



The only possible solution of (1) and (2) is $F_A = F_B = 0$

$\therefore$ Shear stresses are zero: shafts rotate and untorque themselves.

6.10



Equilibrium:

Forces at A & C act through shaft axis

$$\sum M_x = 0 : -M_A - M_C + M_0 = 0 \quad (1)$$

Geometric Compatibility:

$$\phi_{AB} = \phi_{BC} \quad (2)$$

$$M_A = \frac{\phi_{AB}}{L_1} G_1 I_{AB}$$

$$I_{AB} = \frac{\pi}{2} \left(\frac{d_1}{2}\right)^4$$

$$G_1 = G_2$$

$$M_B = \frac{\phi_{BC}}{L_2} G_2 I_{BC}$$

$$I_{BC} = \frac{\pi}{2} \left(\frac{d_2}{2}\right)^4$$

From these, using (1) and (2),

$$M_A = + \frac{M_0}{1 + \frac{L_1}{L_2} \left(\frac{d_2}{d_1}\right)^4} \quad , \quad M_C = + \frac{M_0}{1 + \frac{L_2}{L_1} \left(\frac{d_1}{d_2}\right)^4}$$

(NOTE NEGATIVE SIGN BECAUSE OF ASSUMED DIRECTION)

6.11 As in Problem 6.10,

$$\phi_{AB} = \frac{2 M_A}{\pi G_1} \left(\frac{L_1}{r_1^4}\right)$$

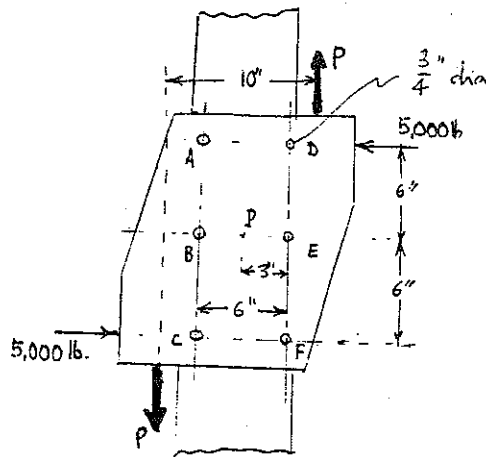
$$, \quad \phi_{AB} = \phi_{BC}$$

$$\phi_{BC} = \frac{2 M_C}{\pi G_2} \left(\frac{L_2}{r_2^4}\right)$$

In addition, $M_A = M_C$

$$\therefore \left(\frac{G_2}{G_1}\right) \left(\frac{L_1}{L_2}\right) \left(\frac{d_2}{d_1}\right)^4 = 1$$

6.12



The set of rivets may be considered to be a non-circular shaft with an externally applied couple, M . From symmetry, any rotation must occur with P as centre. The tangential displacement of any rivet is Ar ,

$A = \text{constant}$

$r = \text{distance from } P$

$\therefore$ Tangential shear stress, assumed constant across rivet $= Br$, $B = \text{const.}$

Moment of this stress about $P = \left(\frac{\pi d^2}{4} r\right)(Br)$, $d = \text{dia. of rivet}$

$\therefore$ Total Moment about $P = \frac{\pi B d^2}{4} \sum r^2 = M$, for equilibrium

$$\therefore B = \frac{M}{\frac{\pi d^2}{4} \sum r^2} \quad \text{Thus } B \propto M.$$

In this problem, $M = 60,000 \text{ lb-in}$, $d = \frac{3}{4}''$, $\sum r^2 = (9+9+45 \times 4) = 198$

$$\therefore B = \frac{60,000}{\frac{\pi}{4} \left(\frac{3}{4}\right)^2 \times 198}$$

Shear stress in A, C, D, F $= B \times \sqrt{45} = \underline{4600 \text{ psi}}$

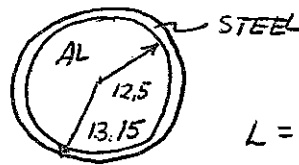
Shear stress in B, E $= B \times 3 = \underline{2050 \text{ psi}}$

Shear stress will reach 10,000 psi first in A, C, D, F. Stress is linearly related to external moment. $\therefore$ Extra external moment permissible is

$$\frac{10,000 - 4600}{4600} \times 60,000 = P \times 10 \quad \therefore P = \underline{7050 \text{ lb}}$$

a) FOR ANY ANGLE OF TWIST

$$\frac{(\tau_{\theta z})_{r_2}}{(\tau_{\theta z})_{r_1}} = \frac{G_{STEEL} r_2}{G_{AL} r_1}$$



$L = 1.5m$

$$\frac{(\tau_{\theta z})_{r_2} / \gamma_{st}}{(\tau_{\theta z})_{r_1} / \gamma_{AL}} = \frac{G_{ST}}{G_{AL}} \frac{\gamma_{AL}}{\gamma_{st}} \frac{r_2}{r_1}$$

DATA STEEL $G_{st} = 82 \text{ GN/m}^2$ $\gamma_{st} \approx 1.38 \text{ GN/m}^2$
 AL $G_{AL} = 26.2 \text{ GN/m}^2$ $\gamma_{AL} \approx 96.5 \text{ MN/m}^2$

$$\frac{(\tau_{\theta z})_{r_2} / \gamma_{st}}{(\tau_{\theta z})_{r_1} / \gamma_{AL}} = 0.241 \quad \text{YIELDING OCCURS FIRST IN AL AT } r=r_1$$

FOR COMPOSITE SHAFTS

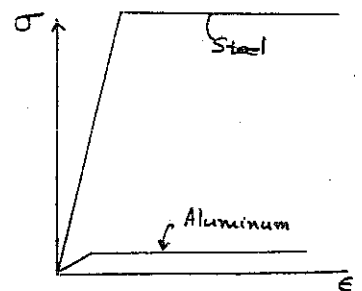
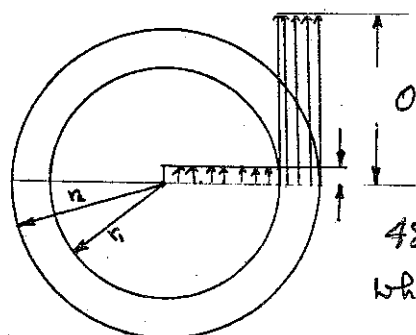
$$(\gamma_{\theta z})_{r_1} = \frac{2M_z G_{AL} r_1}{\pi [G_{AL} r_1^4 + G_{st} (r_2^4 - r_1^4)]} = 0.5 \gamma_{AL} \quad \text{FOR YIELDING USING TRESCA CRITERION}$$

$$\text{SOLVE FOR } M_z = 363 \text{ N}\cdot\text{m}$$

(b) Making the assumption that neither 2040-6 Al alloy nor 4130 steel strain-hardens, the stress-strain curves look like:

at failure, the stress profile

is:



$48 \times 10^{-3} \text{ GN/m}^2$
 When the shaft has yielded completely,

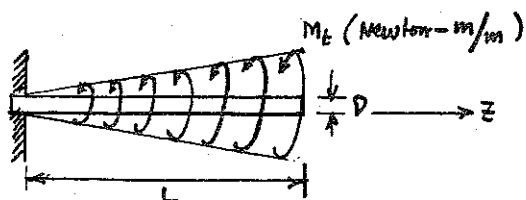
$$M_z = 48 \times 10^{-3} \int_{r_1}^{r_2} 2\pi r^2 dr + 0.69 \int_{r_1}^{r_2} 2\pi r^2 dr = 1130 \text{ N}\cdot\text{m}$$

When plastic deformation commences, $(\tau_{\theta z})_{r_1} = 48 \times 10^{-3} \text{ GN/m}^2$

The relation $\tau_{\theta z} = G r \frac{d\phi}{dz}$ still holds at $r=r_1$

$$\frac{d\phi}{dz} = \frac{\tau_{\theta z}}{G r} = \frac{48 \times 10^{-3}}{26.2 \times 12.5 \times 10^{-3}} = 0.146 \text{ rad/m}$$

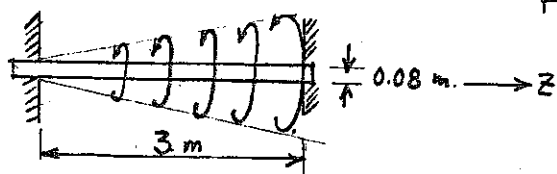
(6-14)



$$I_z = \frac{\pi D^4}{32} ; \text{ at } z, \text{ twisting moment} = M_t z$$

$$\phi = \int_0^L \frac{M_t z}{G I_z} dz = \frac{M_t}{G I_z} \frac{L^2}{2} = \frac{16 M_t L^2}{\pi G D^4}$$

(6-15)



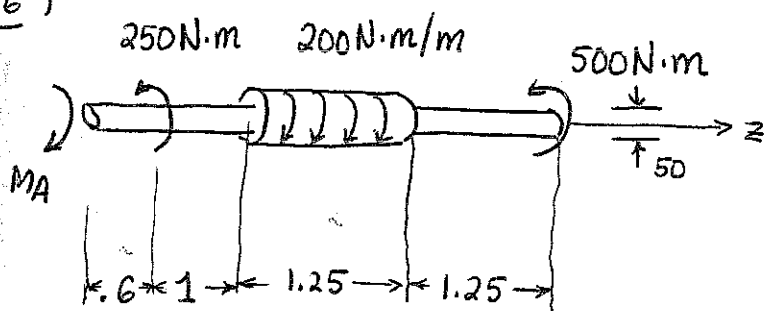
$$\text{For angle of twist at end } \phi = \frac{16 M_t L^2}{\pi G D^4} \text{ (From Prob. 6-14)}$$

$$\text{twisting moment applied at end } M_t = \frac{G I_z \phi}{L}$$

$\therefore$, twisting moment at the walls :

$$= \frac{G I_z}{L} \cdot \frac{M_t L^2}{2 G I_z} = \frac{M_t L}{2} = 1.5 M_t$$

(6-16)



$$\sum M = 0 \quad M_A = 250 + 500 - 200 \times 1.25 = 500 \text{ N}\cdot\text{m}$$

$$\text{ANGLE OF TWIST } \phi = \int_0^z \frac{M_t}{G I_z} dz$$

$$M_t = -500 + 250 \langle z - 0.6 \rangle^0 - 200 \langle z - 1.6 \rangle^1 + 200 \langle z - 2.85 \rangle^1$$

$$I_z = \frac{\pi r^4}{2} \quad \phi = \frac{1}{G I_z} \left[-500z + 250 \langle z - 0.6 \rangle^1 - 100 \{ \langle z - 1.6 \rangle^2 - \langle z - 2.85 \rangle^2 \} \right]$$

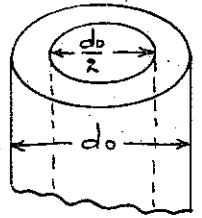
6-17

$$(\tau_{\theta z})_{\max} = \frac{M_t r_o}{I_z} = \frac{M_t (\frac{d_o}{2})}{\frac{\pi}{32} d_o^4 (1 - \frac{1}{16})} = \frac{16 \times 32 M_t}{30 \pi d_o^3}$$

$$M_t = \frac{30 \pi d_o^3 \tau_{\max}}{16 \times 32} \quad 16\text{-ft.}$$

$$\text{Horse-Power} = \frac{N M_t}{33,000}, \quad N = \text{rpm}$$

$$= \frac{\pi N d_o^3 \tau_{\max}}{1100 \times 512} \quad \left[d_o: \text{ft}; \tau_{\max}: \frac{\text{lb}}{\text{ft}^2} \right]$$



6-18

Assume shaft is "thin-walled". Then shear flow does not change with polar angle θ and shear stress is constant across wall.

If q = shear flow,

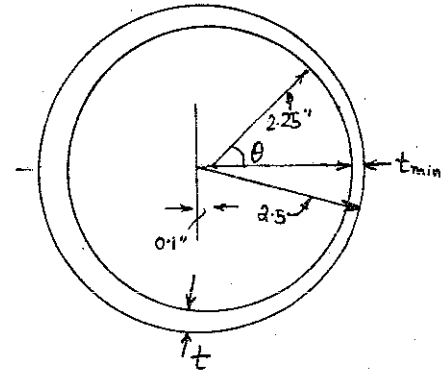
$$M_t \approx 2qA \left\{ \text{where } A = \frac{\pi}{4} \left\{ \frac{5+4.5}{2} \right\}^2 \right\}$$

$$\text{Max. shear stress} = \frac{q}{t_{\min.}}; t_{\min.} = 0.15" \text{ (minimum wall thickness)}$$

$$\text{Max shear stress} = \frac{Y}{2} \text{ for yielding} \approx \frac{35,000}{2} \text{ (Tresca)}$$

$$\text{This gives } q = 0.15 \times \frac{35,000}{2}$$

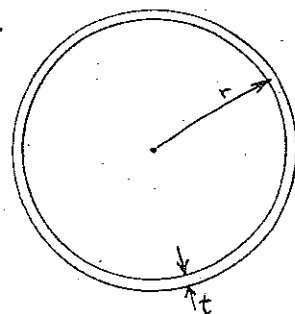
$$(M_t)_{\max} = 0.15 \times 35,000 \times \frac{\pi}{4} \times (4.25)^2 = \underline{\underline{82,000 \text{ lb-in}}}$$



6.19 For a thin-walled tube in general, $I_p \approx 2\pi r^3 t$.

Also, according to equations 6.8 and 6.9,

$$\frac{M_t L}{G I_p} = \frac{\tau}{G} \frac{L}{r}, \quad \tau = \frac{M_t r}{I_p}$$



(a) Yielding starts in outer tube In the outer tube,

$$M_{t_o} = \frac{\tau_y I_p}{2r} = \frac{\tau_y \cdot 2\pi (2r)^3 t}{2r} = 8\pi r^2 t \tau_y$$

In inner tube, $\tau = \frac{1}{2} \tau_y$, $M_{t_i} = \frac{\frac{1}{2} \tau_y \cdot 2\pi r^3 t}{r} = \pi r^2 t \tau_y$

$$\therefore T_y = M_{t_o} + M_{t_i} = 9\pi r^2 t \tau_y \quad \text{and} \quad \phi_y = \frac{M_{t_o} L}{G I_p} = \frac{\tau_y L}{2Gr}$$

(b) Stress in both tubes is τ_y

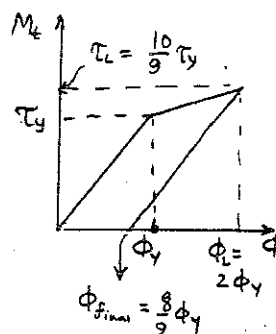
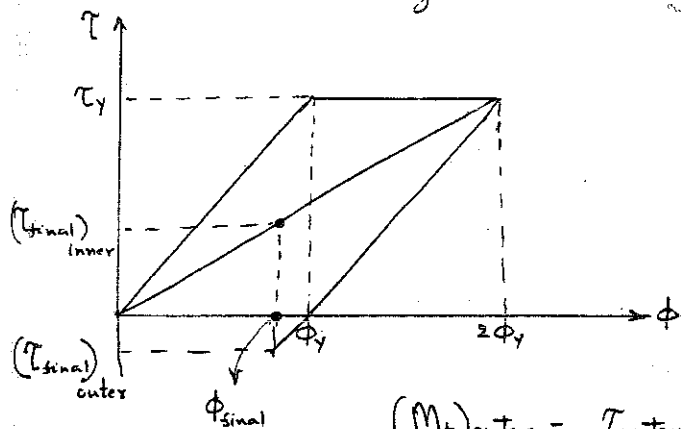
$$M_{t_o} = 8\pi r^2 t \tau_y, \quad M_{t_i} = 2\pi r^2 t \tau_y$$

$$\therefore T_L = 10\pi r^2 t \tau_y$$

(c) Comparing stresses in inner tube, $\phi_i = 2\phi_y = \frac{\tau_y L}{Gr}$

The curve of torque τ , angle for elastic springback is

The stress τ vs twist angle curves are below



$$(M_t)_{outer} = \tau_{outer} \times 4\pi r^2 t$$

$$(M_t)_{inner} = \tau_{inner} \times \pi r^2 t$$

Equilibrium

$$\therefore (M_t)_{outer} + (M_t)_{inner} = 0$$

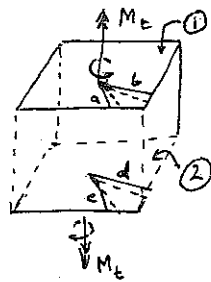
$$\therefore \tau_{outer} = -\frac{1}{4} \tau_{inner}, \text{ for the final stresses}$$

Find from the figure $(\tau_{outer})_{final} = -\frac{1}{9} \tau_y$

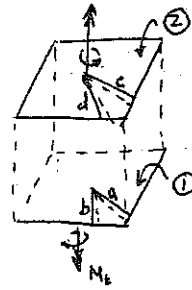
$$(\tau_{inner})_{final} = \frac{4}{9} \tau_y$$

6.20

(a)

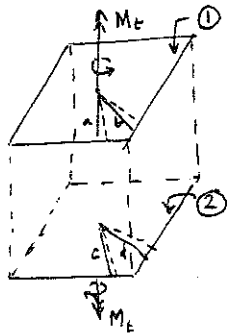


Inverting,

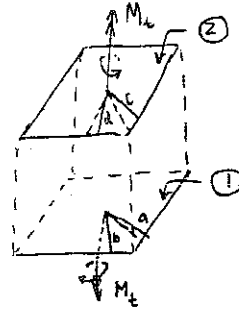


Comparing the upper faces in the two figures, we see that with this deformation, symmetry about diagonals is preserved; by symmetry about diagonals we mean: elements similarly placed with respect to both diagonals undergo similar displacements. This is a geometrically possible mode of deformation.

(b)



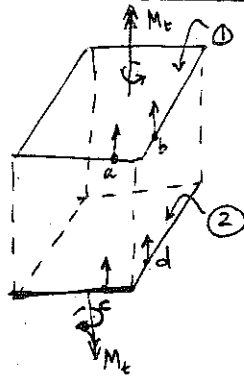
Inverting,



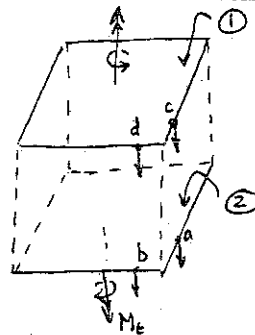
Here, symmetry about diagonals does not exist. This mode of deformation is not geometrically possible.

6.21

(a)

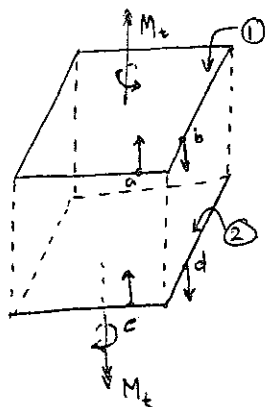


Inverting:-

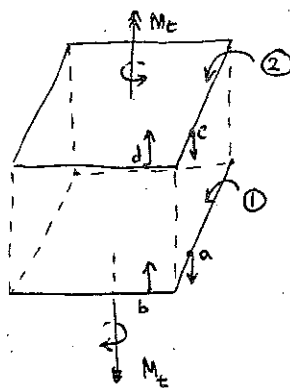


Symmetry about diagonals is not preserved. This is not a geometrically possible mode of deformation.

6.21 cont.
(b)

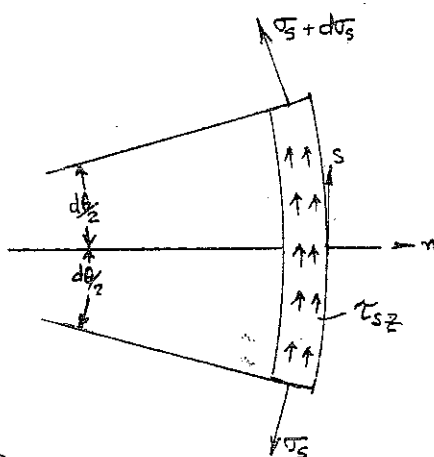
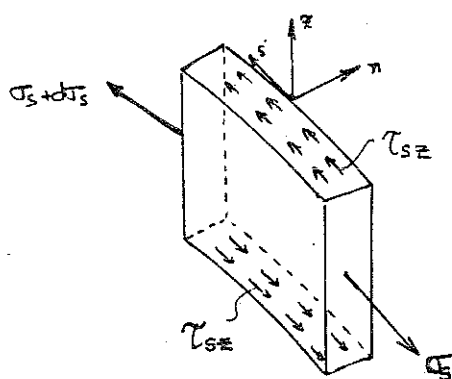


Inverting,



Symmetry about diagonals is preserved. This is a geometrically possible mode of deformation.

6.22



For a shaft with no surface loading,

τ_{sz} on the upper and lower faces must be the same for a given value of the s -coordinate.

$$\text{Equilibrium : } \sum F_n = 0 : \left[\tau_{sz} \frac{d\theta}{2} + (\tau_{sz} + d\tau_{sz}) \frac{d\theta}{2} \right] dndz = 0$$

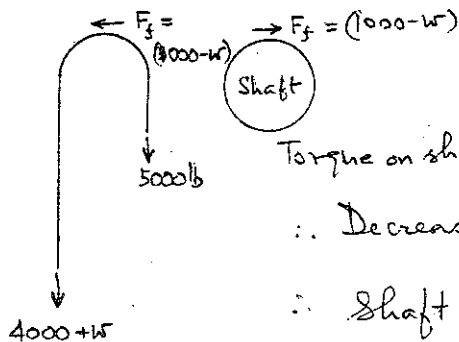
$$\therefore \tau_{sz} = 0$$

6.23 Before passengers:

Torque on shaft

$$M_{ti} = 3000 \text{ lb-ft}$$

After passengers step on (below)



$$\text{Torque on shaft} = (1000 - 3w) \text{ lb-ft}$$

$$\therefore \text{Decrease in Torque} = 3w \text{ lb-ft}$$

$\therefore$ Shaft rotates back through an angle ϕ given by

$$\phi = \frac{M_t L}{G I_z} = \frac{3w \times 12 \times 60}{12 \times 10^6 \times \frac{\pi}{32} \times 2.56} = 7.15w \times 10^{-6} \text{ radians}$$

This lowers the elevator by $\delta_1 = 7.15w \times 10^{-6} \times 36 \text{ inches}$

Due to the increase in tension, the cable elongates by

$$\frac{FL}{AE} = \delta_2 = \frac{w}{(\frac{1}{2})} \times \frac{200 \times 12}{30 \times 10^6} = 13.3w \times 10^{-6} \times 12 \text{ inches}$$

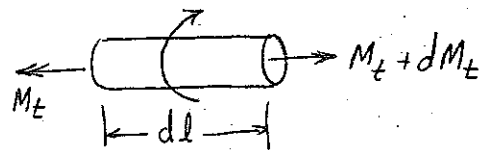
Total downward displacement is

$$\delta = \delta_1 + \delta_2 = 420w \times 10^{-6} \text{ in} = 0.2''$$

$$\therefore \underline{w = 475 \text{ lb}}$$

6.24 AT SOME DISTANCE l FROM THE SWITCH

$$(M_t)_l = M_0 + \int_0^l dM_t$$



WHERE M_0 = TORQUE AT SWITCH = $0.33 \text{ N}\cdot\text{m}$

ASSUMING CONSTANT FRICTION TORQUE ALONG THE CABLE (PER UNIT LENGTH)

$$(M_t)_l = M_0 + \frac{dM_t}{dl} \int_0^l dl = M_0 + l \frac{dM_t}{dl} \quad ; \quad \frac{dM_t}{dl} = 0.45 \text{ N}\cdot\text{m/m}$$

$$\text{TORQUE AT KNOB} = M_0 + 0.45(\text{LENGTH OF CABLE}) = (0.33 + 0.45L) \text{ N}\cdot\text{m}$$

REQUIRE STRESS AT KNOB = τ_y FOR YIELDING

$$(\tau_{\theta z})_{\text{MAX}} \text{ AT KNOB} = \frac{(0.33 + 0.45L) \frac{d}{2}}{I_p} = \frac{0.33 + 0.45L}{\frac{\pi}{16} (3 \times 10^{-3})^3} = \tau_y = 280 \text{ MN/m}^2$$

THIS GIVES $L = 2.55 \text{ m}$

$$\text{PLAY: } \frac{d\phi}{dl} = \frac{(M_t)_l}{GI} = \frac{0.333 + 0.45L}{84 \times 10^9 \times \frac{\pi}{32} \times 81 \times 10^{-12}}$$

$$\phi_L = \int_0^L \frac{d\phi}{dl} dl = \frac{1}{0.668} \left[0.33 \times 2.55 + \frac{0.45}{2} (2.55)^2 \right] = 3.45 \text{ Rad}$$

$$\text{TOTAL PLAY} = 2\phi_L = 6.9 \text{ rad} = 1.1 \text{ REV.}$$

6.25

W = WEIGHT OF WATER ON FOIL

$$\gamma = 9.8 \text{ kN/m}^3$$

$$W = 12 \times 10^{-8} \times 9.8 \times 10^{-3} \text{ N} \\ = 1.17 \times 10^{-9} \text{ N}$$

$$\text{MOMENT ABOUT AB OF } W = 2.34 \times 10^{-8} \text{ N}\cdot\text{mm}$$

M_B = MOMENT APPLIED AT B TO ROTATE DE BACK TO HORIZONTAL

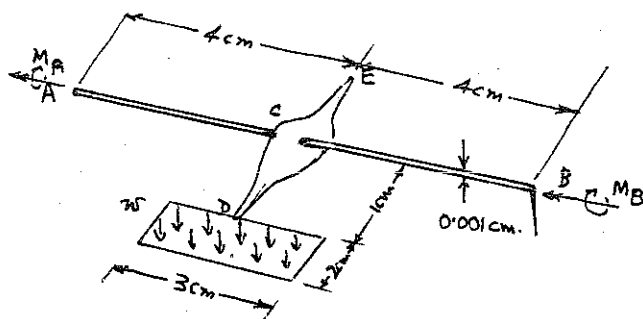
M_A = MOMENT APPLIED AT A BY SUPPORT AT A = ZERO SINCE $\phi_{AC} = 0$

WHEN M_B IS APPLIED TO BRING DE TO HORIZONTAL POSITION

$$\therefore M_B = 2.34 \times 10^{-8} \text{ N}\cdot\text{mm}$$

$$\therefore \phi_B = \frac{2.34 \times 10^{-8} \times 40}{\frac{30 \times 10^9}{10^6} \frac{\pi}{32} 10^{-8}} = \frac{M_t L}{GI_p} \text{ Rads} = 1.82^\circ$$

$$G = 30 \text{ GN/m}^2$$



6.26

For constant shear flow,

$$\tau_{AB} = \tau_{DC} = \frac{1}{2} \tau_{AD} = \frac{1}{2} \tau_{BC},$$

assuming shear stress is constant across wall thickness.

Then, by equation 6.30,

$$M_t = 2Aq_s = 2A \times (t\tau)_{BC}$$

$$= 2 \times [36.4 \times 46.8] \times [1.3]$$

At yielding, $\tau_{BC} = \frac{Y}{2}$ (MAX SHEAR STRESS CRITERION) $\approx \frac{380 \text{ MN/m}^2}{2}$ (Fig 5.28)

$$= 190 \text{ MN/m}^2$$

$$(M_t)_Y = 8.4 \times 10^5 \text{ N}\cdot\text{mm} = 840 \text{ N}\cdot\text{m}$$

(6-27) (a) FOR CONSTANT SHEAR FLOW

$$1.3 \tau_y = \tau_y [2.6 - x] - x \tau_y$$

$$= \tau_y [2.6 - 2x]$$

$$x = 0.65 \text{ mm}$$

(b) $M_t = 2Aq_s$ STILL HOLDS (Eq 6.30)

$$= 2 \times 137 \times 1.3 \times 10^{-3} \times 190 \times 10^6 = 67.7 \text{ N}\cdot\text{m}$$

$$\text{WHERE } A = (52 - 5.2)(39 - 2.6) = 137.3 \text{ mm}^2$$

This is the same as before because the shear flow in the thin wall, and hence everywhere, is the same as before. In reality, $(M_t)_L$ will be somewhat greater than $(M_t)_Y$ because the assumption of constant shear stress across the section at the onset of yielding is incorrect.

6.28 Determine the friction torque about the normal to the ground at the centre of the contact area.

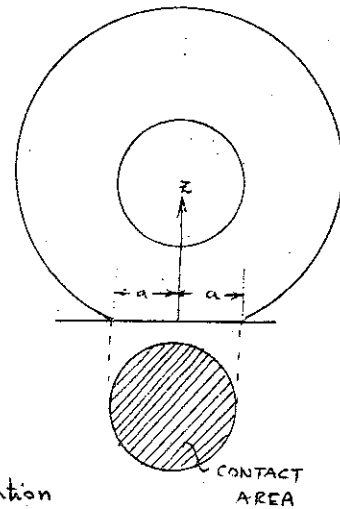
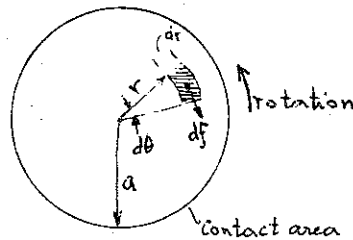
$$\text{Area} = \pi a^2$$

$$\text{pressure} = 30 \text{ psi}$$

$$\therefore \text{Normal load} = 1000 \text{ lb} = 30 \pi a^2$$

$$\therefore \underline{a = 3.25''}$$

When the tire turns, it turns about the central normal to the contact area. On each element of area, a friction force df is exerted by the ground, as shown.



$$df = 0.6 p dA = 0.6 p \times r d\theta dr$$

$$\text{Torque} = df \times r = 0.6 p r^2 d\theta dr$$

$$\begin{aligned} \text{Total Torque} &= \iint_A 0.6 p r^2 d\theta dr = \int_{r=0}^a \int_{\theta=0}^{2\pi} 0.6 p r^2 d\theta dr = 0.6 p \times 2\pi \times \frac{a^3}{3} \\ &= \frac{0.6 \times 30 \times 2\pi \times (3.25)^3}{3} = \underline{1300 \text{ lb-in}} \end{aligned}$$

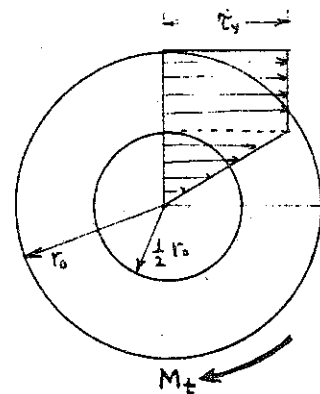
$$\text{Torque due to two wheels} = 2600 \text{ lb-in}$$

$$\text{Torque in steering column (due to 20:1 reduction gear)} = \frac{2600}{20} = 130 \text{ lb-in}$$

$$\tau_{\max} = \frac{M_t \times r}{I_p} = \left(\frac{2 M_t}{\pi r^3} \right) = \frac{2 \times 130}{\pi \times \left(\frac{3}{8} \right)^3} = \underline{1570 \text{ psi}}$$

6.29 Use the method of superposition of p258.

The stress profile in the shaft upon loading is shown alongside. Superpose on this a linear stress a linear reversed stress profile such that its torque is $-M_t$. The assumption



6.29^{cont}

of linearity is good if the max superposed stress τ_m is less than $2\tau_y$, because then the max residual stress is

$$\tau_r = \tau_m - \tau_y \leq \tau_y.$$

The superposed profile is shown alongside.

The initial torque is, from eqns 6.20 and 6.24

$$\phi_i = 2\phi_y$$

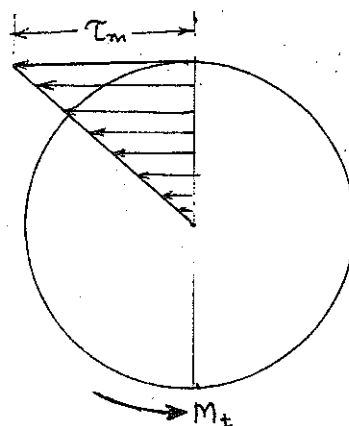
$$M_t = \frac{4}{3}\tau_y \left[1 - \frac{1}{4}\left(\frac{1}{3}\right)^2 \right] = 1.29\tau_y$$

A moment τ_y , elastically applied causes an angular twist ϕ_y .

The superposed elastic moment is $1.29\tau_y$. Therefore the angle of springback is $1.29\phi_y$

$$\therefore \phi_2 = \phi_i - 1.29\phi_y = 0.71\phi_y = \underline{\underline{0.355\phi_i}}$$

[Note: $\tau_m < 2\tau_y$ because superposed Torque = $1.29\tau_y < 2\tau_y$].

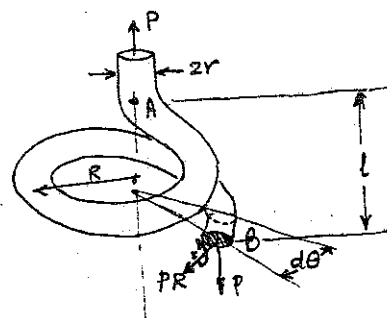


6.30

Cut the spring at some point B, making the cut perpendicular to the wire. If the pitch is small, the plane of the cut will be almost parallel to the axis of the spring, and for equilibrium we require, at B, a force P and a moment PR , $\perp$ to the axis.

The moment PR causes twisting of an element of the wire of length $R d\theta$ through an angle $\Delta\phi = \frac{PR \cdot (R d\theta)}{GI_p} = \frac{2}{\pi} \frac{PR^2}{Gr^4} d\theta$

Now consider the lower part of the spring (not shown) to remain fixed and determine the effect at A due solely to the twist angle $\Delta\phi$: A rotates about B through $\Delta\phi$, in the plane of the section at B. The upward component is $d\delta = R\Delta\phi$. The sideward



6.30 cont.

component is $R \Delta\phi$, but when we integrate this over the whole spring, it vanishes from symmetry. Integrating the upward component over the whole spring,

$$\begin{aligned} \delta &= \int_0^L d\delta = \int_0^L R \Delta\phi = R \frac{2}{\pi} \cdot \frac{PR^2}{Gr^4} \cdot 2\pi n \\ &= \frac{4PR^3n}{Gr^4} \quad \text{where } n = \text{no. of coils.} \end{aligned}$$

6.31 ESTIMATES: TWISTING TORQUE $\approx 0.565 \text{ N}\cdot\text{m} \approx 5 \text{ LB}\cdot\text{IN}$
 COMPRESSIVE FORCE $\approx 70 \text{ N} \approx 15 \text{ lb}$

$$\text{COMPRESSIVE STRESS} = \frac{F}{A} = \frac{F}{\frac{\pi d^2}{4}} = \frac{4 \times 70 \text{ N}}{\pi \times 36 \times 10^{-6}} = 2.5 \text{ MN/m}^2$$

$$\text{SHEAR STRESS} = \frac{M_t / (\frac{d}{2})}{\frac{\pi}{32} d^4} = \frac{0.56 \text{ N}\cdot\text{m} \times 3 \times 10^{-3}}{\frac{\pi}{32} (16)^4 \times 10^{-12}} = 13.2 \text{ MN/m}^2$$

$$(\text{COMPRESSIVE STRESS} / \text{SHEAR STRESS}) = 0.19$$

6.32 (a) ASSUME LOADING IS ENTIRELY

ELASTIC. SINCE 1100-O + 2024-T4 HAVE SAME ELASTIC MODULUS, WE CAN WRITE

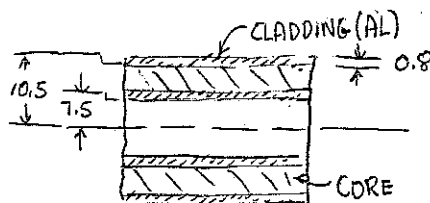
$$\tau = \frac{M_t r}{I_z}$$

$$I_z = \int_{6.7}^{11.3} r^2 dA = 2\pi \int r^3 dr = \frac{\pi}{2} [(11.3)^4 - (6.7)^4] = 2.24 \times 10^4 \text{ mm}^4$$

$$\text{at INSIDE RADIUS} \quad \tau = 400 \times 0.3 \times 6.7 \times 10^{-3} / 2.24 \times 10^{-8} = 35.8 \text{ MN/m}^2$$

τ_y for 1100-O $\approx 17.5 \frac{\text{MN}}{\text{m}^2}$ (Fig 5.5b), and therefore the inner layer of cladding will yield. Similarly the outer layer will yield. Therefore the assumption of elastic loading is wrong.

(b) Assume both inner & outer cladding become plastic but core



6.32 CONT

REMAINS ELASTIC; AS SHOWN IN FIG(1b)

M_c = MOMENT DUE TO THE CLADDING

$$= 17.5 \times 2\pi \times 6.7 \times 0.8 \times 6.7 \\ + 17.5 \times 2\pi \times 10.5 \times 0.8 \times 10.5 = 13.65 \text{ N}\cdot\text{m}$$

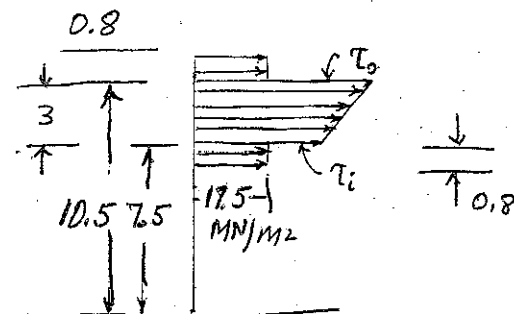
APPLIED MOMENT = $400 \times 0.3 = 120 \text{ N}\cdot\text{m}$

MOMENT IN CORE = $120 - 13.65 = 106.35 \text{ N}\cdot\text{m}$

$$(I_2)_{\text{CORE}} = 2\pi \int_{7.5}^{10.5} r^3 dr = 14.123 \times 10^3 \text{ mm}^4$$

$$\tau_c = \frac{106.35 \times 7.5 \times 10^{-3}}{14.123 \times 10^{-9}} = 56.5 \text{ MN/m}^2$$

$$\tau_o = 7.91 \text{ MN/m}^2$$

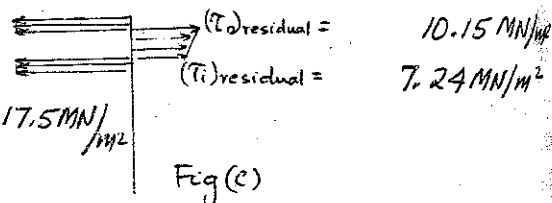


- (c) To obtain the residual stresses, we add to the stresses calculated above the ^{ELASTIC} stresses due to a moment of $120 \text{ N}\cdot\text{m}$ in the reverse direction. From the calculation in (a), we see that this reverse moment will just cause the inner cladding to yield in the reverse direction; and, of course, cause greater yielding in the outer cladding. The resulting stress distribution will therefore be as in Fig(c). The residual moment due to the cladding is $13.65 \text{ N}\cdot\text{m}$, and this must be balanced by a $13.65 \text{ N}\cdot\text{m}$ moment due to the core.

The stresses in the core are:

$$(\tau_i)_{\text{residual}} = \frac{13.65 \times 7.5 \times 10^{-3}}{14.123 \times 10^{-9}} = 7.24 \text{ MN/m}^2$$

$$(\tau_o)_{\text{residual}} = 10.148 \text{ MN/m}^2$$



6.33

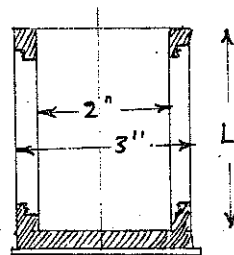
Let depth of hole = L ft.

Neglect buoyancy due to water

Weight of metal supported

$$\approx 12L \times \frac{\pi}{4} [3^2 - 2^2] \rho \quad \text{where } \rho = \frac{16}{173}$$

$$= 12L \times \frac{\pi}{4} \times 5 \times 0.28 = 13.2L \text{ lb.}$$



$\therefore F$ (see Fig) = $13.2L$ lb. Assuming uniform tensile stress across section,

6.33 cont

tensile stress $\sigma_z = \frac{13.2 L}{\frac{\pi}{4} (3^2 - 2^2)} = 3.36 L \text{ psi}$

Shear stress: $\tau_{\theta z} = \frac{M_z r_o}{I_z} = \frac{2000 \times 12 \times \frac{1}{2}}{\frac{\pi}{32} [3^4 - 2^4]} = 5650 \text{ psi}$

In the radial direction, there is a stress due to water pressure, but it is negligible where σ_z is largest, at the top.

Mohr's Circle:

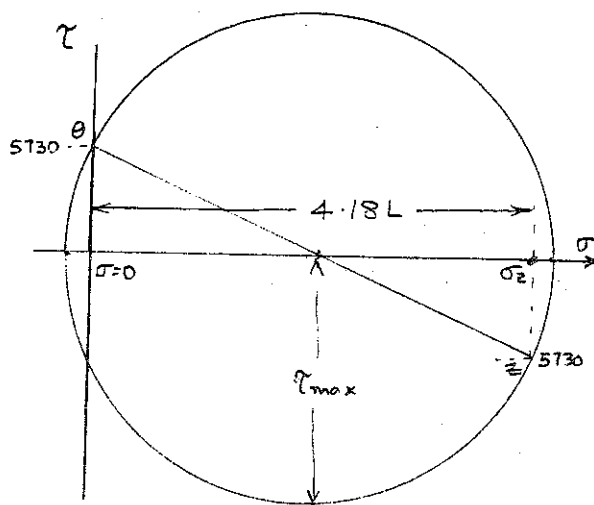
$$\tau_{max} = \left\{ (5650)^2 + \left(\frac{3.36 L}{2} \right)^2 \right\}^{\frac{1}{2}}$$

Using max. shear stress criterion, we must have

$$\tau_{max} = \frac{Y}{2} \approx 25,000 \text{ psi}$$

$$\left\{ (5650)^2 + (1.68 L)^2 \right\}^{\frac{1}{2}} \approx 25,000$$

This gives $L \approx 14,500 \text{ ft}$



6.34 The only fixed requirement is (2): spring constant = $125 \frac{\text{lb}}{\text{in.}}$

Torque = $P \cdot x$; windup angle $\phi = \frac{(P \cdot x) L}{G I} = \frac{P x L}{G \frac{\pi}{32} d^4}$

where d = wire dia.

Deflection at load = ϕx . Then by definition,

spring constant $k = \frac{P}{\phi x} = \frac{G \pi d^4}{32 x^2 L} = \frac{12 \times 10^6 \pi}{32} \frac{d^4}{x^2 L}$

or $\frac{d^4}{x^2 L} = \frac{32 \times 125}{12 \times 10^6 \pi} = \frac{1}{3000 \pi} \dots \dots \dots (1)$

The load is composed of static: 1000 lb

dynamic $(\pm 6'' \times 125 \frac{\text{lb}}{\text{in}}) = \pm 750 \text{ lb}$

max load = $1000 + 750 = 1750 \text{ lb.}$

$(M_t)_{max} = 1750 x \text{ lb-in}$

6-21

6.34 cont.

$$\tau = \frac{(1750x) \frac{d}{2}}{\frac{\pi}{32} d^4} = 8900 \frac{x}{d^3}$$

we require

$$\frac{x}{d^3} \cdot 8900 \leq 50,000, \text{ or } \frac{x}{d^3} \leq 5.62 \dots \dots (2)$$

Equations (1) and (2) can be satisfied by many combinations of x , L and d . One possibility is obtained by taking for x and L their maximum permissible values of 30" and 120".

Then from (1) $d^4 = \frac{900 \times 120}{3000\pi} = 11.4$ and $d = \underline{1.84"} \underline{\hspace{1cm}}$

Equation (2) is satisfied: $\frac{x}{d^3} = \frac{30}{6.12} = 4.85 < 5.62$

$$\tau = \frac{4.85}{5.62} \times 50,000 = 43,000 \text{ psi}$$

Alternatively, (1) can be written $Ld^2 = \frac{3000\pi}{x^2/d^6}$

and substituting (2), $Ld^2 \geq \frac{3000\pi}{(5.62)^2}$

Ld^2 is proportional to the volume of the shaft and a cost design would make this a minimum:

$$Ld^2 = \frac{3000\pi}{(5.62)^2}$$

Taking $L = 120"$ as a further design condition, we get

$$d = 1.57" \quad x = 21.4"$$

Taking $x = 30"$ as another alternative, we get

$$d = 1.75" \quad L = 98.0"$$

In both these cases, $\tau = 50,000 \text{ psi}$.

6.35

Consider Free-Body of PE and EQ:

For equilibrium,

$$M_P = W \frac{a}{2} - a F_E \quad (1)$$

$$M_Q = F_E \cdot a \quad (2)$$

Geometric Compatibility

Displacement at E of 'C' = Displacement at E of 'D'

$\therefore \phi_P$ (rotation of shaft at P) = ϕ_Q (rotation of shaft at Q)

$$\phi_P = \frac{2M_P}{\pi G} \frac{L}{\left(\frac{d_A}{2}\right)^4} \quad \phi_Q = \frac{2M_Q}{\pi G} \frac{L}{\left(\frac{d_B}{2}\right)^4}$$

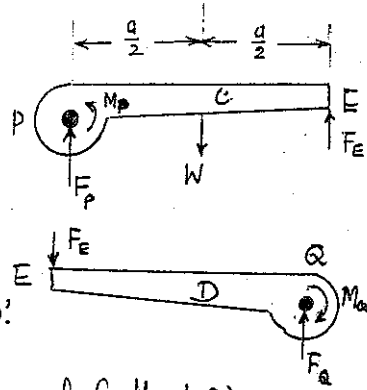
$$\therefore \frac{M_P}{d_A^4} = \frac{M_Q}{d_B^4} \quad (3)$$

From (1), (2) and (3)

$$M_P = \frac{W a}{2 \left[1 + \left(\frac{d_B}{d_A} \right)^4 \right]}, \quad M_Q = \frac{W a}{2 \left[1 + \left(\frac{d_A}{d_B} \right)^4 \right]}$$

$$F_E = \frac{W}{2 \left[1 + \left(\frac{d_A}{d_B} \right)^4 \right]}$$

$$\phi_P = \phi_Q = \frac{16 W a L}{\pi [d_A^4 + d_B^4] G}$$



6-36

Expressions in 4.19 are $\epsilon_r = \frac{\partial u}{\partial r}$ $\epsilon_\theta = \frac{1}{r} \frac{\partial v}{\partial \theta} + \frac{u}{r}$
 $\epsilon_z = \frac{\partial w}{\partial z}$ $\gamma_{r\theta} = \frac{\partial v}{\partial r} + \frac{1}{r} \frac{\partial u}{\partial \theta} - \frac{v}{r}$
 $\gamma_{\theta z} = \frac{1}{r} \frac{\partial w}{\partial \theta} + \frac{\partial v}{\partial z}$ $\gamma_{rz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial r}$

We have $u=0$, $v=r\phi$, $w=0$, giving

$$\epsilon_r = \epsilon_\theta = \epsilon_z = \gamma_{r\theta} = \gamma_{rz} = 0 \text{ and } \gamma_{\theta z} = r \frac{d\phi}{dz}$$

Stokes' Law:

$$\epsilon_r = \frac{1}{E} [\sigma_r - \nu(\sigma_\theta + \sigma_z)] \quad \epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)]$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\epsilon_r + \epsilon_\theta)] \quad \gamma_{r\theta} = \frac{1}{G} \tau_{r\theta}; \quad \gamma_{\theta z} = \frac{1}{G} \tau_{\theta z}; \quad \gamma_{rz} = \frac{1}{G} \tau_{rz}$$

From Section 6.3, $\sigma_r = \sigma_\theta = \sigma_z = \tau_{r\theta} = \tau_{rz} = 0$

$$\therefore \epsilon_r = \epsilon_\theta = \epsilon_z = \gamma_{r\theta} = \gamma_{rz} = 0$$

$$\text{and } \tau_{\theta z} = G \gamma_{\theta z} = G r \frac{d\phi}{dz}$$

Equilibrium Equations $\frac{\partial \tau_{\theta z}}{\partial z} = 0$ is satisfied because $\frac{d\phi}{dz} = \frac{M_z}{GI_z} \neq f(z)$.

$$\frac{1}{r} \frac{\partial \tau_{\theta z}}{\partial \theta} = 0 \text{ is satisfied because } \tau_{\theta z} \neq f(\theta).$$

The first equilibrium eqn does not contain $\tau_{\theta z}$ and is identically satisfied.

6-37

From Symmetry, $\gamma_{r\theta} = \gamma_{rz} = 0$; $\frac{\partial}{\partial \theta} = 0$; $\gamma_{\theta z} = r \frac{d\phi}{dz}$

Assume $\epsilon_r = \epsilon_\theta = 0$.

Invoking Stokes' law, we can get:

$$\sigma_r = \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_r + \epsilon_\theta + \epsilon_z) + \frac{E \epsilon_r}{1+\nu}$$

$$\sigma_\theta = \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_r + \epsilon_\theta + \epsilon_z) + \frac{E \epsilon_\theta}{1+\nu}$$

$$\sigma_z = \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_r + \epsilon_\theta + \epsilon_z) + \frac{E \epsilon_z}{1+\nu}$$

$$\text{In this problem, } \sigma_r = \frac{\nu E \epsilon_z}{(1+\nu)(1-2\nu)} = \sigma_\theta, \quad \sigma_z = \frac{(1-\nu)E}{(1+\nu)(1-2\nu)},$$

$$\tau_{r\theta} = \tau_{rz} = 0, \quad \tau_{\theta z} = G r \frac{d\phi}{dz}$$

Substituting in the equilibrium equations, prob 4.4, we get

$$\frac{\nu E}{(1+\nu)(1-2\nu)} \frac{\partial \epsilon_z}{\partial r} = 0, \quad G r \frac{d^2 \phi}{dz^2} = 0, \quad \frac{(1-\nu)E}{(1+\nu)(1-2\nu)} \frac{\partial \epsilon_z}{\partial z} = 0$$

From symmetry, $\frac{\partial \epsilon_z}{\partial \theta} = 0 \therefore \epsilon_z = \text{constant}$

$$\therefore \sigma_z = \frac{(1-\nu)E}{(1+\nu)(1-2\nu)} \epsilon_z = \text{constant.}$$

But at the surface, with no loading, $\sigma_z = 0 \therefore \epsilon_z = 0$ everywhere.

(6-38) DIAMETER OF SPECIMEN = 1.9 mm

RESULTS OF 4 TESTS : BREAKING TORQUES 0.1, 0.125, 0.1, 0.11 N.m

AVERAGE BREAKING TORQUE 0.1088 N.m

$$\gamma_{\theta z} = \frac{M_t r}{I_z} = \frac{2 M_t}{\pi r^3} ; r = 0.95 \text{ mm}$$

$$\tau_{\theta z} = 81 \text{ MN/m}^2$$

(6-39) Strain energy when $(\tau_{\theta z})_{\max} = (\tau_{\theta z})_{\text{yield}}$

$$U'_S = \frac{1}{2} \int \frac{M_t^2}{GI} dz = \frac{1}{2} \frac{(\tau_{\theta z})_{\text{yield}}^2 I_z^2 L}{GI_z R^2} ; M_t = \frac{I_z \tau_{\theta z}}{R} \quad R = \text{radius of spring (shaft)}$$

$$= \frac{1}{2} \frac{(\tau_{\theta z})_{\text{yield}}^2 I_z L}{GR^2} \quad I_z = \frac{\pi R^4}{2} \rightarrow U'_S = \frac{(\tau_{\theta z})_{\text{yield}}^2 \pi R^2 L}{4G}$$

Strain-energy when all material at $(\tau_{\theta z})_{\text{yield}}$

$$U_S = \frac{1}{2} \int_V \frac{(\tau_{\theta z})^2}{G} dv = \frac{1}{2} \int_V \frac{(\tau_{\theta z})_{\text{yield}}^2}{G} dv = \frac{(\tau_{\theta z})_{\text{yield}}^2 \pi R^2 L}{2G}$$

$$\therefore \text{Efficiency} = \frac{U'_S}{U_S} = \frac{(\tau_{\theta z})_{\text{yield}}^2 \pi R^2 L}{4G} \cdot \frac{2G}{(\tau_{\theta z})_{\text{yield}}^2 \pi R^2 L} = 50\%$$

6.40

Eqn (6.2) remains $\gamma_{\theta z} = r \frac{d\phi}{dz}$

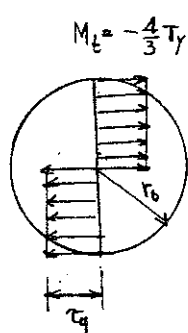
Eqn (6.3) becomes $k r^2 = r \frac{d\phi}{dz}$

$$M = \int r \tau dA = \int r \cdot 2\pi r \left(\frac{r}{k}\right)^{\frac{1}{2}} \left(\frac{d\phi}{dz}\right)^{\frac{1}{2}} dr = \sqrt{\frac{d\phi}{dz}} \cdot \frac{4\pi r^{\frac{7}{2}}}{7k^{\frac{1}{2}}}$$

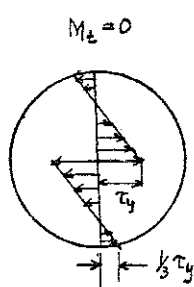
$$\therefore \frac{d\phi}{dz} = \frac{49 M^2 k}{16 \pi^2 r^7} \quad \text{and} \quad \tau = \frac{7M}{4\pi r^3}$$

6-25

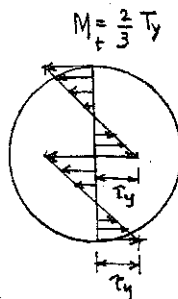
(6-41)



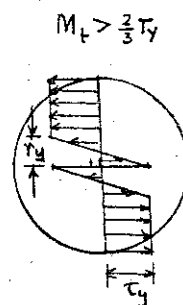
(a) Pre-twist



(b) Released Shaft
(Residue Stress)



(c) yield-point



(d) Plastic deformation

Assume that pre-twist (taken as negative) to be in plastic region

From eq. (6-26) $M_t = -\frac{4}{3} T_y$ $\tau_{\theta z} = \tau_y$ (yield stress) $\phi = -\frac{4}{3} \phi_y$

The shaft is released, (fig. b), and there is a residue stress distribution.

Now apply a positive twisting moment, the shaft behaves elastically until the situation (c)

In Elastic region, stress distribution:

$$\tau_{\theta z} = -\tau_y + G(\Delta r_{\theta z}) = -\tau_y + \frac{Gr}{L} \left[\phi - \left(-\frac{4}{3} \phi_y \right) \right]$$

$$\textcircled{1} \rightarrow \frac{\tau_{\theta z}}{\tau_y} = -1 + \frac{r}{r_0} \left(\frac{\phi}{\phi_y} + \frac{4}{3} \right) \quad \text{where} \quad \frac{Gr_0}{L} \phi_y = \tau_y$$

Solving for $r=r_y$ at which transition into plastic region takes place

$$r=r_y \quad \text{at} \quad \tau_{\theta z} = \tau_y$$

$$\text{From } \textcircled{1} \text{ we get } \frac{r_y}{r_0} = \frac{2}{\phi/\phi_y + 4/3} \quad \textcircled{2}$$

In Plastic Region.

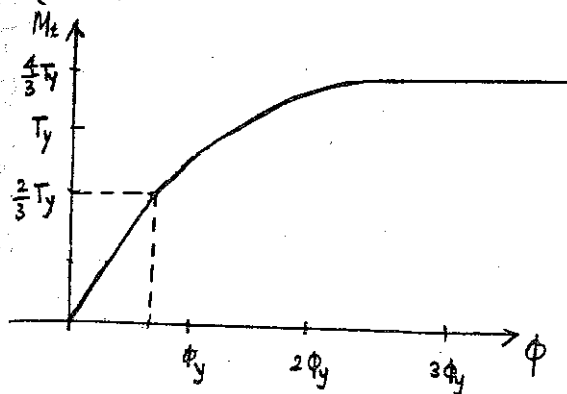
$$\tau_{\theta z} = \tau_y$$

$$\begin{aligned} \therefore \text{Total twisting moment } M_t &= \int_A r \tau_{\theta z} dA \\ &= \int_0^{r_y} r \tau_y \left[-1 + \frac{r}{r_0} \left(\frac{\phi}{\phi_y} + \frac{4}{3} \right) \right] 2\pi r dr + \int_{r_y}^{r_0} r \tau_y 2\pi r dr \end{aligned}$$

Carrying out integration and using $\textcircled{2}$

$$\frac{M_t}{T_y} = \frac{4}{3} - \frac{2}{3} \left(\frac{2}{\phi/\phi_y + 4/3} \right)^3 \quad T_y = \frac{\pi}{2} \tau_y r_0^3$$

(6-41) (cont'd)



stress-distribution at onset of yielding
fig (c). :

$$\tau_{\theta z} = -\tau_y + \frac{2r}{r_o} \tau_y$$

Moment distribution at yield point:

$$\begin{aligned} M_t &= \int_A r \tau_{\theta z} dA = \int_0^{r_o} r \left(-\tau_y + \frac{2r}{r_o} \tau_y \right) 2\pi r dr \\ &= \frac{2}{3} \left(\frac{\pi}{2} \tau_y r_o^3 \right) = \frac{2}{3} T_y \end{aligned}$$

$\therefore$ the shaft behaves elastically up to

$M_t = \frac{2}{3} T_y$ and the curve is linear
up to this point

(6-42) (See Fig. 6-28, 6-29)

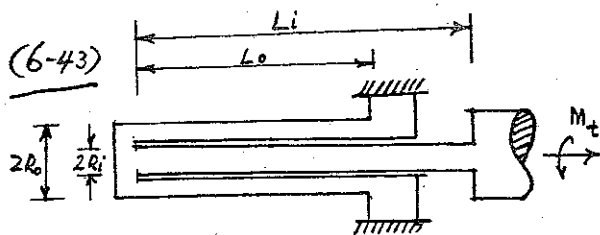
The strain energy for torsion $U_t = \frac{1}{2} \iiint \tau_{\theta z} \gamma_{\theta z} dV$ (6.12)

$$= \frac{1}{2G} \iiint \tau_{s_z}^2 dndsdz$$

$$\gamma = \tau_{s_z} t \rightarrow U_t = \frac{1}{2G} \iint \frac{q^2}{t} ds dz$$

From eq. (6.34) $q = \frac{M_t}{2A} \therefore U_t = \frac{M_t^2 L}{8A^2 G} \oint \frac{ds}{t}$

$\therefore$, Angle of twist $\phi = \frac{\partial U_t}{\partial M_t} = \frac{M_t L}{4A^2 G} \oint \frac{ds}{t}$



$$\phi = \frac{M_t L}{G I_p}$$

Twisting moment is the same in both shaft

(a)

$$\phi_{\text{total}} = \phi_i + \phi_o = \frac{M_t L_i}{G I_{p_i}} + \frac{M_t L_o}{G I_{p_o}} = \frac{M_t}{G} \left\{ \frac{L_i}{I_{p_i}} + \frac{L_o}{I_{p_o}} \right\} = \frac{M_t}{G} \left\{ \frac{L_i I_{p_o} + L_o I_{p_i}}{I_{p_i} I_{p_o}} \right\}$$

$$\text{Torsional spring constant} = \frac{M_t}{\phi_{\text{total}}} = \frac{G I_{p_i} I_{p_o}}{L_i I_{p_o} + L_o I_{p_i}}$$

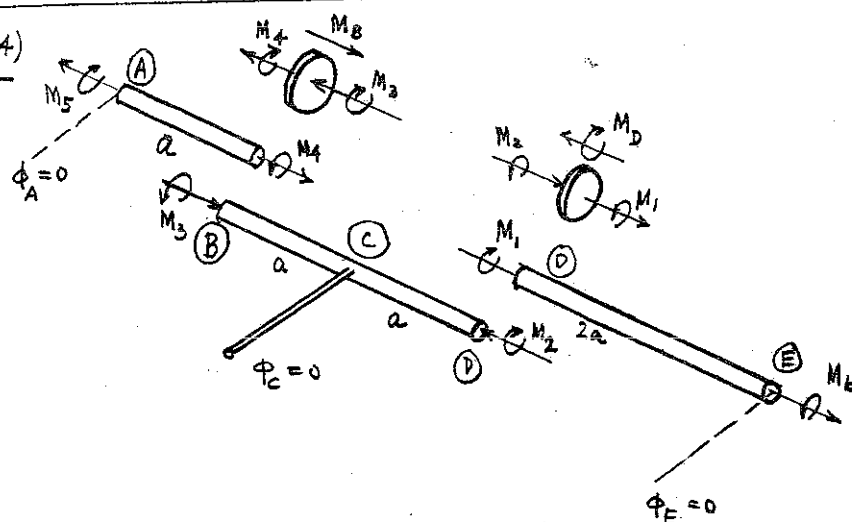
(b) Let maximum permissible stress = τ_y

$$\tau_{y_i} = \frac{M_t R_i}{I_{p_i}} \quad \tau_{y_o} = \frac{M_t R_o}{I_{p_o}}$$

$$\text{For } \tau_{y_i} = \tau_{y_o} \rightarrow \frac{R_i}{I_{p_i}} = \frac{R_o}{I_{p_o}} \rightarrow \frac{R_o}{R_i} = \frac{I_{p_o}}{I_{p_i}} = \frac{\frac{\pi R_o^4}{2} - \frac{\pi R_i^4}{2}}{\frac{\pi R_i^4}{2}}$$

$$\therefore \frac{R_o^4}{R_i^4} - \frac{R_o}{R_i} - 1 = 0$$

(6-44)



From Geometry

$$\phi_A = 0 \quad (1)$$

$$\phi_C = 0 \quad (2)$$

$$\phi_E = 0 \quad (3)$$

As a consequence of (1) & (2)

$$\phi_{BA} = \phi_{BC} \quad (4)$$

As a consequence of (2) & (3)

$$\phi_{DC} = \phi_{DE} \quad (5)$$

From equilibrium:

$$\left. \begin{aligned} M_1 &= Wb \\ M_D &= M_1 + M_2 \\ M_3 &= M_2 \\ M_B &= M_3 + M_4 \end{aligned} \right\} (6)$$

From Force-deformation:

$$\left. \begin{aligned} \phi_{BA} &= \frac{M_4 a}{G I_z} & \phi_{DC} &= \frac{M_2 a}{G I_z} \\ \phi_{BC} &= \frac{M_3 a}{G I_z} & \phi_{DE} &= \frac{M_1 2a}{G I_z} \end{aligned} \right\} (7)$$

Solving (4), (5), (6), (7) we get $M_1 = Wb$, $M_2 = 2Wb$, $M_3 = 2Wb$, $M_4 = 2Wb$

$$\therefore M_B = 4Wb \quad M_D = 3Wb$$

(6-45) Stress due to cannon ball = $\frac{10 \text{ Kg}}{(2.54)^2 \times 10^{-6} \text{ m}^2}$; corresponding strain $\epsilon = \frac{\sigma}{E}$

Force at each end = $F = \frac{G m_1 m_2}{d^2} = \frac{6.67 \times 10^{-11} \times 5 \times 100}{(0.4)^2}$
 $= 2.08 \times 10^{-7} \text{ newton}$

$$= \frac{10}{1.4 \times 10^9 (2.54)^2 \times 10^{-6}}$$

$$= 1.15 \times 10^{-3}$$

$\therefore$ The Mylar tape can support the load.

Total Torque = $M_t = F \times 2m = 4.16 \times 10^{-7} \text{ newton-m}$

Angular deflection $\phi = \frac{G_2 M_t L}{a b^3 G}$

$a = 25.4 \text{ mm} = 2.54 \times 10^{-2} \text{ m}$
 $b = 0.254 \text{ mm} = 2.54 \times 10^{-4} \text{ m}$ } $\frac{a}{b} = 10$

From Table 6.1, $\frac{a}{b} = 10$ $G_2 = 0.312$

$G = \frac{E}{2(1+\nu)} = \frac{E}{2.8} = \frac{1.4 \times 10^9 \text{ N/m}^2}{2.8} = 5 \times 10^8 \text{ N/m}^2$

$L = 10 \text{ m}$

$M_t = 4.16 \times 10^{-7} \text{ N-m}$

$\therefore \phi = \frac{0.312 \times 4.16 \times 10^{-7} \times 10}{2.54 \times 10^{-2} \times (2.54)^3 \times 10^{-12} \times 5 \times 10^8} \text{ radians}$
 $= 7.5 \times 10^{-2} \text{ radians}$

$\therefore$ At 1 m away, deflection = 7.5 cm.

$\therefore$ Cannon balls deflect 7.5 cm. ; ITERATION MAY BE CARRIED OUT.